EduVision_DV

Dashboard Planning & Prototyping — Storyboard

Purpose: Define the four dashboard layouts, filters, navigation, dashboard actions, and interactive comparisons before dashboard development.

Dashboard	Primary Purpose	Core Views	Primary Filters
University Overview	High-level university performance	KPI cards, global distribution, top universities, performance comparison	Country, University, Ranking Range, Region
Research Analytics	Research impact and productivity analysis	Research impact, citations, productivity, research comparison	Country, University, Research KPI Range
Student Analytics	Student population and internationalization	International students, faculty-to-student ratio, enrollment comparison	Country, University, Student KPI Range
Country Comparison	Country-level education benchmarking	Country map, ranking comparison, regional trends, KPI benchmarking	Country, Region, KPI Range

Design principle: Consistent header, filter area, KPI area, analytical visuals, and navigation across all dashboards. The prototype should demonstrate the interaction flow before full dashboard development.

1. University Overview

High-level view of university performance, rankings, global distribution, academic reputation, and institutional comparison.

HEADER EduVision_DV title + dashboard name + navigation controls
FILTER BAR Country | University | Ranking Range | Region
KPI CARDS Total Universities | Average/Selected Global Ranking Score | Research Impact | Academic Reputation
GLOBAL DISTRIBUTION World map showing university distribution
TOP UNIVERSITY RANKINGS Ranked university table with University, Country, Global Ranking Score
PERFORMANCE COMPARISON Interactive comparison of Global Ranking Score vs Research Impact Score
FOOTER / NAVIGATION University Overview | Research Analytics | Student Analytics | Country Comparison

Interactive behavior: Selecting a country or university filters the analytical views. Selecting a university highlights it in comparison visuals. Navigation controls move between the four dashboards.

2. Research Analytics

Focus on research quality, research environment, citation performance, and research productivity.

HEADER Dashboard title + navigation controls
FILTER BAR Country | University | Research KPI Range
KPI CARDS Research Impact Score | Research Productivity Index | Research Quality | Citation Score
RESEARCH IMPACT University comparison using Research Impact Score
CITATION PERFORMANCE Citation performance by university
RESEARCH PRODUCTIVITY Research Productivity Index comparison
RESEARCH RELATIONSHIP Research Environment vs Research Quality comparison
FOOTER / NAVIGATION University Overview | Research Analytics | Student Analytics | Country Comparison

Interactive behavior: University selection highlights the selected institution across research views. Country filtering updates all research comparisons.

3. Student Analytics

Analyze student population, international student participation, faculty-to-student ratio, and student distribution.

HEADER
Dashboard title + navigation controls

FILTER BAR
Country | University | Student KPI Range

KPI CARDS
Student Population | International Student % | Faculty-to-Student Ratio | University Count

INTERNATIONAL STUDENTS
International student percentage comparison

FACULTY-TO-STUDENT RATIO
University ratio comparison

ENROLLMENT COMPARISON
Student population by university / country

STUDENT DISTRIBUTION
Distribution of student-related indicators

FOOTER / NAVIGATION
University Overview | Research Analytics | Student Analytics | Country Comparison

Interactive behavior: Country and university selections update student metrics and comparisons. Users can compare selected institutions against the filtered population.

4. Country Comparison

Compare country-level higher-education performance using university coverage and aggregated KPI measures.

HEADER Dashboard title + navigation controls
FILTER BAR Country | Region | KPI Range
KPI CARDS Universities by Country | Average Global Ranking | Average Research Impact | Average Academic Reputation
COUNTRY MAP Geographic distribution and country-level performance
COUNTRY RANKING Country comparison table / chart
REGIONAL COMPARISON Regional education performance comparison
KPI BENCHMARKING Country-level KPI comparison across selected countries
FOOTER / NAVIGATION University Overview | Research Analytics | Student Analytics | Country Comparison

Interactive behavior: Selecting a country updates country-level KPI views and linked university analysis. Multiple selected countries can be compared.

5. Interaction & Navigation Specification

These behaviors form the prototype acceptance criteria for Module 4.

Feature	Specification
Global filters	Country, University, and relevant KPI/range filters should affect the views on the active dashboard.
Dashboard navigation	Navigation controls provide direct movement among University Overview, Research Analytics, Student Analytics, and Country Comparison.
University selection	Selecting a university highlights or filters related visualizations on the dashboard.
Country selection	Selecting a country filters the relevant university and country-level views.
Interactive comparisons	Selected institutions/countries can be compared against the current filtered population.
KPI consistency	KPI labels and definitions remain consistent across dashboards.

Prototype acceptance checklist

- ■ All four dashboard layouts represented.
- ■ Filters defined for each dashboard.
- ■ Navigation path defined across all dashboards.
- ■ Dashboard actions and interactive comparisons defined.
- ■ Layout is consistent enough to translate into Tableau.
- ■ Prototype can be reviewed before full dashboard development.

Planned Module 4 deliverables: dashboard_storyboard.pdf and eduvision_prototype.twbx